HS-OBS	Functional Requirements Specification
DEPARTMENT	DOCUMENT TYPE

VZI R1

Operating Modes Functional Requirements Specification
External doors and sliding steps

NONDISCLOSURE CLAUSE

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Rev	Date	Author	Revision Description
00	21/07/2023	G. Tondini	First Issue
01	27/10/2023	G. Tondini	Revised section: 2.1 System Configuration Swapped the colour lamp (Red / White) on LIP description. 3.1.2.2.3 Green Loop Added filter requirement for to show the train door status on IDU 3.1.6.2.2 PRM door closing Deleted requirement : 2F_14.Zefiro-Europe.CONCEPT.768 2F_14.Zefiro-Europe.CONCEPT.770 2F_14.Zefiro-Europe.CONCEPT.772 3.2.4.2 Requirements Add close PB in req 2F_14.Zefiro-Europe.CONCEPT.838 3.2.5.2 Requirements Add green light color in req 2F_14.Zefiro-Europe.CONCEPT.336 3.2.6.1.1Normal condition and 3.2.6.2 Requirements Swapped the colour lamp (Red / White) on LIP description. 3.2.8.2 Requirements Added the requirement for manage the door out of service indication on IDU (removing the status data for the door isolated)
02	07/10/2024		Removed requirements : 2F_14.Zefiro-Europe.CONCEPT.368 2F_14.Zefiro-Europe.CONCEPT.369

Table of contents

1 Introduction 5
1.1 Train Configuration 5
1.2 Definitions and abbreviations 5
2 System Overview 6
2.1 System Configuration 6
3 Functional requirements 8
3.1 Exterior Doors: Overall Door Control 8
3.1.1 Door Release 8
3.1.2 Close External Doors 14
3.1.3 Step operation 18
3.1.4 Door Status during Cab activation/deactivation 20
3.1.5 Supervision and fault handling 20
3.1.6 Special functions for PRM door 25
3.2 Local Door Control 28
3.2.1 Cab doors 28
3.2.2 Open external door 29
3.2.3 Close external door 30
3.2.4 Door status indication 34
3.2.5 Local crew operation 35
3.2.6 Status indication of each door 39
3.2.7 Obstacle Detection 40
3.2.8 Isolation of Doors 45
3.2.9 Emergency opening exterior doors 48
3.2.10 Train unit coupling and uncoupling 53
4 Signal Interface 54
4.1 I/O Interface 54
4.1.1 Supplier system 54
4.1.2 TCMS SW 56
4.2 Bus Interface 56
5 References 57

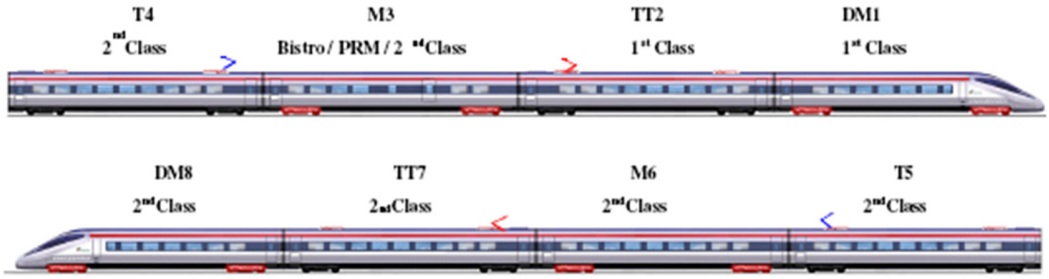
List of Tables

1 Introduction

This document is a function specification for the control of the exterior doors and step system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
TST PRM	2008/164/EC – COMMISSION DECISION OF 21ST February 2008 concerning a technical specification of interoperability relating to 'persons with reduced mobility' in the trans-European conventional and high speed system
PRM	Persons with Reduced Mobility
RIC-KEY	Key according to the agreement on mutual use of vehicles for persons and goods in the international transport
DCU	Door Control Unit
TL	Train Line

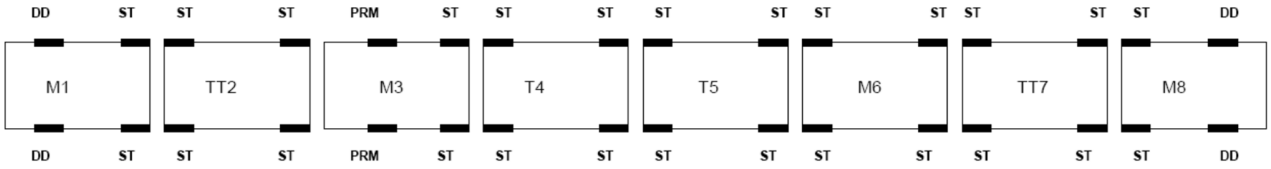
2 System Overview

2.1 System Configuration

The passenger external doors system is made of:

- 26 passenger standard external doors, half left opening and half right opening and the doors control system at train level. Each door is equipped with a DCU. Standard doors are equipped with a mobile step
- 2 single PRM doors. M3 car is also equipped with a couple of external PRM doors, one on each side, both right opening. These doors can be locally operated only by the train crew and have the exclusive use to allow the access and the egress of wheelchair users though the relevant lift; no other passengers shall be allowed to use these doors. The lift is considered a separate component and not a part of the PRM door.

The train is also equipped with four driver's doors, one on each side of the DM1 and DM8 cars, manually operated.



ST = Passengers standard door

PRM = PRM door for wheel chair users

DD = Drivers door

MAN-MACHINE INTERFACES

Interfaces in driver desk

- Closing push buttons
- Right doors side release push button
- Left doors side release push button
- Released right side doors signal light
- Released left side doors signal light
- Open doors signal light
- IDU (for selective door de-activation)
- Crew interfaces at each standard door
- Internal emergency opening command (by RIC key operation)
- External emergency opening command RIC key operated
- Internal local door isolation device

- External local door isolation device
- Three positions crew switch for central closing command and closing command reset
- Signal lamp related to three positions crew switch
- Crew interface at train staff compartment
- IDU (for selective door de-activation)

Passengers interfaces at each standard door

- Internal local opening command
- Internal local closing command
- External opening local command
- Internal emergency opening command (by breaking the sealing)
- Warning buzzer
- Door status panel LIP (yellow = not released, green= released, white ~~red~~ = door or step isolated, red ~~white~~ = door and step isolated)
- Visual warning lamp

Crew interfaces at each PRM door

- Local door isolation device
- Internal three position open/close crew switch RIC key operated
- External three position open/close crew switch RIC key operated
- Signal lamp (indicating that the door is released)
- Manual release device inside
- Visual Warning lamp

Maintenance interface at each DCU

- Service push button

HARDWIRED SIGNALS

1) Doors release signal 1
2) Doors release signal 2
3) Zero speed signal
4) Speed >=10 km/h signal
~~5)~~ ~~PRM door coding~~
5) PRM Lift in parking position
6) Reset central close
7) DCL loop

3 Functional requirements

3.1 Exterior Doors: Overall Door Control

3.1.1 Door Release

3.1.1.1 Function design

The doors are kept closed and locked until a releasing command is issued.

The door releasing command is available from the driver desk by means of two push buttons:

- Left door release push button,
- Right door release push button

Released door may be open by the local push buttons.

3.1.1.1.1 Normal condition

Starting state: doors are not released, closed and locked and a cab is active

The door release command is effective only from the driver desk of the occupied cab, at standstill and no general closing command is issued in the same time by the train staff.

The releasing command is effective for only one side of the train.

Releasing of doors of both sides of the train is possible by two releasing commands, one for each side

The door release command is by a couple of hardwire train lines for each train side.

Operation of the door release push button shall not be effective when the train is not at standstill or from the driver desk of a not occupied cab or when a general closing command is present. In such cases the voltage level of the right/left door release hardwire train line do not change.

In washing mode door release shall not be possible.

After a door releasing command has been operated the related door release TL will remain at the active voltage level up to the train is no longer at standstill or a general closing command is given by the train staff of by the driver.

When the doors on one side of the train are released the related push button on the driver desk is lighted.

In addition to the door release TL, the releasing of the doors for local opening is depending among the following others hardwired signals:

- Zero speed (Train at standstill = active train line)
- Speed >=10 km/h (Speed ≥10 km/h = active train line)

In normal condition the door is released when the following combination of states of signals is present:

- Door release 1: Active
- Door release 2: Active
- Zero speed: Active
- Speed >=10 km/h: not active

When a door is released for local opening an acoustic signal is given that is clearly audible to persons inside the train.
This alert signal will sound for a minimum of five seconds unless the local opening push button is operated, in which case it ceases after 3 seconds.
In addition, visual indication is given using the led lights on door command push buttons.
The operation of the local opening push button of a released door starts the opening sequence of the door.
In the event of loss of power, the door remains in the actual state, that is if the door is open, the door remains open, if the door Is closed and locked, the door remains closed and locked.

3.1.1.1.2 **Failure cases**

Wrong state of the release train lines.
In addition to the state of signals for releasing in normal condition, to improve the availability of the doors, in case of failure of one of the release signals, the door is released for opening also when the following states of the release signals are present.
Note: The zero speed signal and the speed>=10km/h is not allowed to be faulty, this is solved in HW to obtain signal not affected by single faults.

Door release1	**Not active**	Active
Door release 2	Active	**Not active**
Zero speed	Active	Active
Speed>=10 km/h	Not active	Not active

For all other combinations of the above four signals the doors are not released for local opening.
To obtain full independence from door release signals and zero speed signal, activation of the door release TLs have to be dependent from the standstill state of the train and not from the state of the zero speed train line.

3.1.1.1.3 **Failure detection**

The DCU makes available to TCMS the detected state of the four hardwired signals (mirrored signal status).
The incoherence among the TLs state and the mirrored signals from DCU is detected by TCMS for diagnostic purpose.
In addition, the state of the HW input signal is compared with the corresponding SW input signal (see § 3.1.5) for internal diagnostic purpose.

3.1.1.1.4 **Failure handling**

Single failure of input HW input signals does not affect the normal operation of the doors. The failure has to be removed during maintenance.
Double failure of input HW input signals does not allows releasing of the doors of one train side. The train has to take away from the service. Opening of the doors is possible by emergency devices.

3.1.1.2 Requirements

3.1.1.2.1 Release exterior door by driver

ID	2F_14.Zefiro-Europe.TRS.64
Description	Permission to open exterior doors shall be activated by the driver by pushing the door release push buttons.
Safety level	SIL 2

ID	2F_14.Zefiro-Europe.TRS.222
Description	It shall only be possible to release the doors when the train speed is 0 km/h.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.269	The two doors release trainlines shall be set on the right side only when the driver presses the door release PB on the right side and the train speed is zero and the cab is active.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
2F_14.Zefiro-Europe.CONCEPT.270	The two doors release trainlines shall be set on the left side only when the driver presses the door release PB on the left side and the train speed is zero and the cab is active.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2

ID	2F_14.Zefiro-Europe.TRS.175
Description	The doors shall be kept closed and locked until the driver releases them.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.274	Local opening command shall not be active when both doors release trainlines are not active.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2

ID	2F_14.Zefiro-Europe.TRS.180
Description	The doors shall only be possible to open when the speed is equal to 0 km/h.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.275	The door shall only be possible to open when the conditions according to table below are fullfilled: Door release 1: A | NoA | A Door release 2: A | A | NoA Zero speed: A | A | A Speed >= 10 km/h: NoA | NoA | NoA A=Active NoA=Not Active	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2

ID	2F_14.Zefiro-Europe.TRS.177
Description	Door release shall be cancelled automatically when train speed > 0 km/h
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.279	When the train speed gets not zero the doors release trainlines shall be deactivated	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_14.Zefiro-Europe.CONCEPT.786	When the train speed is not zero and zero speed relay based on BCU signal states zero and zero speed relay based on WSP signal does not state zero, then the doors release trainlines 1 of both sides shall be deactivated, in order to prevent an unwanted release.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_14.Zefiro-Europe.CONCEPT.787	When the train speed is not zero and zero speed relay based on BCU signal states zero and zero speed relay based on WSP signal does not state zero, an event shall be set to inform the driver about the recovery of unwanted release and consequent reduction of the availability of this function.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_14.Zefiro-Europe.CONCEPT.788	When the train speed is not zero and zero speed relay based on WSP signal states zero and zero speed relay based on BCU signal does not state zero, then the doors release trainlines 2 of both sides shall be deactivated, in order to prevent an unwanted release.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_14.Zefiro-Europe.CONCEPT.789	When the train speed is not zero and zero speed relay based on WSP signal states zero and zero speed relay based on BCU signal does not state zero, an event shall be set to inform the driver about the recovery of unwanted release and consequent reduction of the availability of this function.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_14.Zefiro-Europe.TRS.223
Description	Door release shall be cancelled when close request is given by crew using one local panel.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.280	The DCU shall transmit a closing command to the train control unit when a close request is given by crew using one local panel.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.281	The train control unit shall deactivate all the Doors release trainlines when a closing command is received from the DCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_14.Zefiro-Europe.TRS.224
Description	Door release shall be cancelled when close request is given by driver.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.282	The train control unit shall deactivate all the Doors release trainlines when the closing PB on the driver desk of the active cab is operated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_14.Zefiro-Europe.TRS.302
Description	In washing mode door release shall not be possible.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.774	The train control unit shall deactivate all the Doors release trainlines when washing mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_14.Zefiro-Europe.TRS.174
Description	In the event of loss of power to the door controls, the doors shall be kept locked by the locking mechanism.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.283	If DCU loses power supply, when the door is closed and locked, the door shall be kept closed and locked	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

3.1.1.2.2 Side Selection for Door Release

ID	2F_14.Zefiro-Europe.TRS.66
.Description	Permission shall be given individually for each side (side selective) of the train -> 2 push-buttons (1x for left side, 1x for right side).
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.285	When the train speed is zero, and the cab is active, and the open PB on the right side of the drivers desk is pushed the doors release trainlines on the right side shall be activated	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

ID	2F_14.Zefiro-Europe.TRS.311
Description	Undue Door Release Command (several doors are unlocked (with train crew not correctly informed of this door status) or released or opened in inappropriate areas or situations) shall have a risk controlled to an acceptable level.
Safety level	SIL 3

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.790	each contact of the right side open PB shall be connected to a relay (dalayed in opening): - one NO contact of each relay shall be read by the TCMS safe - one NO contact of each relay shall be used as criteria for the trainline activation - one DO of the TCMS safe shall be used as criteria for trainline activation	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.792	when the right side 'opening' relay (see req.790) is read (due to PB pressed) by the TCMS the DO inhibition shall be closed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_14.Zefiro-Europe.CONCEPT.794	the safe DO inhibition for any side (right side and left side) shall be given (DO open) when the speed >3 km/h	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

2F_14.Zefiro-Europe.CONCEPT.286	When the train speed is zero and the cab is active and the open PB on the left side of the drivers desk is pushed the doors release trainlines on the left side shall be activated	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.791	each contact of the left side open PB shall be connected to a relay (dalayed in opening): - one NO contact of each relay shall be read by the TCMS safe - one NO contact of each relay shall be used as criteria for the trainline activation - one DO of the TCMS safe shall be used as criteria for trainline activation	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.793	when the left side 'opening' relay (see req.791) is read (due to PB pressed) by the TCMS the DO inhibition shall be closed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_14.Zefiro-Europe.CONCEPT.781	When the contact of open PB which activates the door release trainlines 1 is stuck closed (i.e. more than 30 s – as default of adjustable preset time – TCMS reads the position as pressed), the doors release trainlines 1 of both sides shall be deactivated, in order to prevent an unwanted release. This close command shall be kept until the mentioned contact is considered stuck closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_14.Zefiro-Europe.CONCEPT.782	When the contact of open PB which activates the door release trainlines 1 is stuck closed (see previous req.), an event shall be set to inform the driver about the recovery of unwanted release and consequent reduction of the availability of this function.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_14.Zefiro-Europe.CONCEPT.783	When the contact of open PB which activates the door release trainlines 2 is stuck closed (i.e. more than 30 s – as default of adjustable preset time – TCMS reads the position as pressed), the doors release trainlines 2 of both sides shall be deactivated, in order to prevent an unwanted release. This close command shall be kept until the mentioned contact is considered stuck closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_14.Zefiro-Europe.CONCEPT.784	When the contact of open PB which activates the door release trainlines 2 is stuck closed (see previous req.), an event shall be set to inform the driver about the recovery of unwanted release and consequent reduction of the availability of this function.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_14.Zefiro-Europe.CONCEPT.785	In case of fault combinations that can involve to loss of both release train lines of each side, the TCMS close command can't be generated automatically but following a close command issued by the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_14.Zefiro-Europe.TRS.225
Description	The push button in the cab shall be lighted when the doors are released
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.287	When at least one of the door release trainlines on the right side is active, the PB on the right side of the driver desk shall be illuminated	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 0
2F_14.Zefiro-Europe.CONCEPT.725	When at least one of the door release trainlines on the right side is active, the PB on the right side of the driver desk shall be illuminated	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

2F_14.Zefiro-Europe.CONCEPT.288	When at least one of the door release trainlines on the left side is active, the PB on the left side of the driver desk shall be illuminated	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 0
2F_14.Zefiro-Europe.CONCEPT.726	When at least one of the door release trainlines on the left side is active, the PB on the left side of the driver desk shall be illuminated	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

3.1.1.2.3 Acoustic Signal for Door Release

ID	2F_14.Zefiro-Europe.TRS.166
Description	When the doors are released a signal, different from the audible door closing alarm, shall be given that is clearly audible to persons inside the train. This alert signal shall sound for a minimum of five seconds unless the door is operated, in which case it may cease after 3 seconds.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.289	When the door is released, DCU shall energize the relevant buzzer to produce an audible release warning . The audible warning shall last for: · 3 seconds if the local open PB is operated within 3 seconds from releasing, · 5 seconds if the door is not opened, · Until operation of the open PB if it is operated between 3 and 5 seconds from releasing	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.729	Via SW parameters, it shall be possible to adjust the maximum time of release warning.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

3.1.2 Close External Doors

3.1.2.1 Function design

Doors close because local closing command, acting only on the local door, or because general closing commands, acting on all the doors of the train.

The function considered in this paragraph is the general closing from the driver desk.

On the driver desk there are two close buttons with the functionality to close all the doors of the train.

A specific input signal to the DCU is not provided. The closing command is performed by deactivation of both the release TLs.

3.1.2.1.1 Normal condition

Starting state: doors are released on one or both sides and a cab is active

In the active cab if at least one door is open the light of the close push buttons is on.

When TCMS detects the operation of one of the close push buttons of the active cab, the TCMS de-activate all the release TLs of the train (both sides of the train).

When the door release TLs are de-activated, the DCU of the not closed doors starts the normal closing sequence. When both the release TLs related to one side of the train become not active the light of the open push button of the related side on the driver desk is turned off.

The normal closing sequence is done by the following steps:

- ♣ The acoustic and visual lamp pre-warning signal is activated,
- ♣ After 3 s the door leaf starts to close,
- ♣ When the door is in closed and locked position, the step starts to close and the acoustic and visual warning lamp signal is de-activated when both door and step are closed and locked,
- ♣ When the door and the step are closed the section of the green loop related to the door is closed,

During the closing sequence the obstruction detection in active.
In addition, open doors close with the normal closing sequence if the train speed becomes >0 km/h.
When all doors are locked and the related movable steps are closed the green loop related to the train become closed.
When the green loop related to the train become closed the door train control system turns off the light of the close push buttons on the driver desk.
When the green loop related to the train is closed the traction of the train becomes allowed.
By closing an open sealed switch, traction of the train becomes possible with the train green loop open.

3.1.2.1.2 **Failure cases**

a) one of the release train line is not deactivated when a general closing command is given
b) one door does not close
c) the green loop is not active when all doors are closed.

3.1.2.1.3 **Failure detection**

a) The TCMS monitors the state of the train line at both end of the train-unit and detects the incoherent states of the lines.
b) The TCMS receives from the not closed door the state of the door and related step. The door or the related step has to be isolated in accordance with the diagnostic message.
c) The TCMS receives from all the DCU the state of the doors. If the TCMS receives SW signals of closed state from all the doors of the train and the green loop is not active a diagnostic message is set from TCMS.

3.1.2.1.4 **Failure handling**

a) Single failure of input HW input signals does not affect the normal operation of the doors. The failure has to be removed during maintenance.
b) The door or the related step has to be isolated in accordance with the diagnostic message.
c) The green loop has to be by-passed by operation of the sealed switch.

3.1.2.2 Requirements

3.1.2.2.1 Close external doors by driver command

ID	2F_14.Zefiro-Europe.TRS.187
Description	A general close command shall be issued if the door close push button in the cab is pressed.
Safety level	SIL 0

ID	2F_14.Zefiro-Europe.TRS.188
Description	When pressing the door close push button in the cab the release signal for both sides shall be withdrawn.
Safety level	SIL 0

ID	2F_14.Zefiro-Europe.TRS.219
Description	There shall be one close pushbutton closing all doors of the train.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.290	When the door close PB on the drivers desk is pressed and the cab is active the doors release trainlines of both sides shall be deactivated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_14.Zefiro-Europe.CONCEPT.727	When the door close PB on the drivers desk is pressed and the cab is active the doors release trainlines of both sides shall be deactivated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_14.Zefiro-Europe.TRS.189
Description	A general closing command shall override any other command.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.291	When the relase signal is withdrawn, the opening command from the local open push buttons shall not be active	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.825	When the release signal is withdrawn, if the local open push button is depressed DCU shall energize the relevant red led on door open PB (steady signal) for at least 1 second or until the PB is depressed. (NOT ACCEPTED INPUT FUNCTION)	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.292	When the relase signal is withdrawn, the closing command from the local close push buttons shall not be active	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.826	When the release signal is withdrawn, if the local closing push button is depressed DCU shall energize the relevant red led on door close PB (steady signal) for at least 1 second or until the PB is depressed. (NOT ACCEPTED INPUT FUNCTION)	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

ID	2F_14.Zefiro-Europe.TRS.217
Description	The visual indication that the door is released shall cease when the door release has been withdrawn. 3 seconds after the door release has been withdrawn the door shall start to close.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.293	When the doors release trainlines get deactivated,, the door released PB visual indication shall cease .	Derived Requirement	4S_12.-.Exterior Door Systems	HW HMI	N/A
2F_14.Zefiro-Europe.CONCEPT.294	When the doors release trainlines get deactivated and the zero speed signal is still active, the door shall start to close after 3 seconds.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.730	Via SW parameters, it shall be possible to adjust the time between deactivation of the release TL and starting of door closing movement.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

ID	2F_14.Zefiro-Europe.TRS.213
Description	All doors shall automatically close when speed > 0 km/h is detected.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.531	When the train speed is >0 km/h, the DCU shall start immediately the acoustic warning signal for door closing.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.731	After 2 s train speed is >0 km/h, the DCU shall start to close the door leaf.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.533	When the door leaf is closed and locked the DCU shall close the step	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.532	The acoustic warning signal shall last until the door leaf and the step are closed and locked.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

3.1.2.2.2 **Audible warning**

ID	2F_14.Zefiro-Europe.TRS.49
Description	An audible warning sound at each door shall be activated 3s before the doors start to close and continue until the door is closed. The sound shall be different in tone to that used when the door is released.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.295	When the doors release trainline is deactivated, the DCU shall energize the local buzzer to produce an audible closing warning. The warning shall last until the door leaf and the step are closed. The warning signal is not activated in case of closing by service push button.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

3.1.2.2.3 **Green Loop**

ID	2F_14.Zefiro-Europe.TRS.136
Description	All doors must be closed and locked in order for the train to depart.
Safety level	SIL 2

ID	2F_14.Zefiro-Europe.TRS.298
Description	It shall only be possible to drive the train if the on-board lift is stowed.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.296	When the door and the related step are closed and locked the DCU shall close the local green loop interface (chain of closed contacts).	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.297	The Green Loop shall be activated when the local green loop interface of all doors are closed.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.298	The train shall only be allowed to depart if the green loop is active, i.e all doors are closed and locked. Information: The PRM door is not possible to close unless lift are in stowed position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.537	By operation of a sealed switch, it shall be possible to start the train when the green loop is not active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.538	The TCMS shall detect when the sealed switch is operated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_14.Zefiro-Europe.TRS.191
Description	The closed status of all doors and steps shall be indicated to the driver by a lamp in driver's desk.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.299	When the green loop is not active the indication lamp on the driver desk shall be illuminated.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
	The TCMS shall apply a proper filter in order to correctly manage the Green loop status. This filter has to be forseen for manage correctly the following inormation manage from TCM : • door status - green loop HW signal • door status received form all the DCU via ETH At least 3 sec of delay for acquire all these signals has to be forseen before to show the train door status on IDU				

3.1.3 Step operation

3.1.3.1 Function Design

The step operation is coordinated with the door leaf operation.

The step moves only when the door leaf is closed. The door leaf moves only when the step is fully open.

3.1.3.1.1 Normal condition

When a door open command is given, the step starts to open immediately. When the step is full open the door leaf is unlocked and starts to open.

A door open command is effective only if the door is released, that is the zero speed signal and the door release signal are active.

3.1.3.1.2 Failure cases

a) The step does not open
b) The step open partially
c) The step does not close

3.1.3.1.3 Failure detection

a) The DCU detects the step is in closed position after an opening command and sets a diagnostic message
b) If the open step has a depth of at least 150 mm corresponding to the minimum allowed by TSI PRM, the DCU accepts the position as open position and the door leaf open subsequently, other ways the door leaf remains closed.
c) The DCU detects the step is in open position after a closing command and set a diagnostic message after number of closing attempts

3.1.3.1.4 Failure handling

a) The step and the door have to be isolated.
b) If the door leaf open, no action are required, otherwise, if the non-opening of the step is not due to the presence of an obstacle, the door has to be isolated.
c) The step has to be manually closed and the step and the door have to be reset after removing the obstacle or be isolated.

3.1.3.2 Requirements

ID	2F_14.Zefiro-Europe.TRS.139
Description	The steps shall only extend if the train is at standstill.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.300	The movement of the step in open direction shall be possible only if the train speed is 0 km/h.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.827	The extension of the step must be completed before opening the door	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1

3.1.4 Door Status during Cab activation/deactivation

3.1.4.1 Function description

The state of the single door system is not affected by the activation or de-activation of the cab.

3.1.4.1.1 Normal conditon

When a cab is activated or deactivated, the state of the door release TLs and of the Zero speed TL remains unchanged. The open and close push buttons of the not active cabs are not active.

3.1.4.1.2 Failure case

a) The state of one or more enable train lines change from de-activate to activate or vice versa when the cab changes its state and the open push button on the driver desk are not activated.

3.1.4.1.3 Failure detection

If the control of the release train line is independent among them, a coherence check allows to detect the state change due to failure.

3.1.4.1.4 Failure handling

To be removed during maintenance.

3.1.4.2 Requirements

ID	2F_14.Zefiro-Europe.TRS.53
Description	When a cab is deactivated, each door (including step) status shall remain until a cab is activated again and driver gives a new release/close order.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.301	When the cab is deactivated the current status of the door release trainlines and zero speed signal shall remain.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

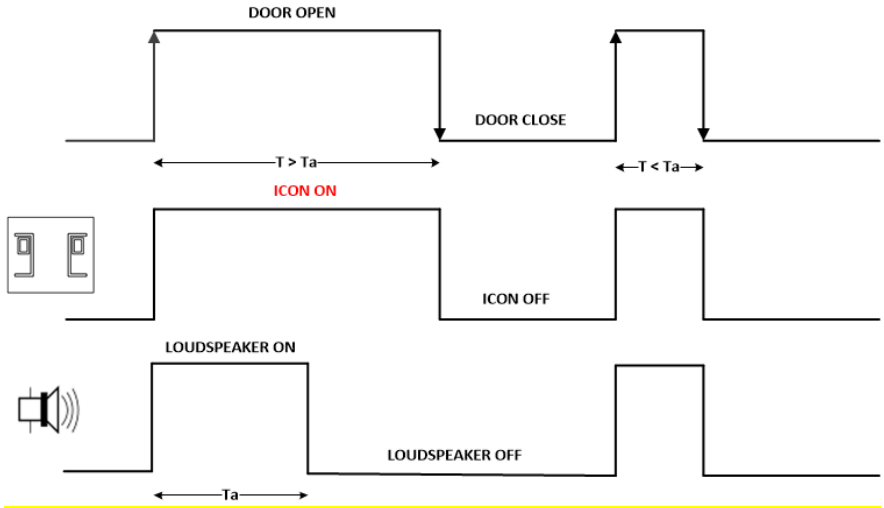
3.1.5 Supervision and fault handling

3.1.5.1 Function design

In normal condition the check of coherence of the input and output signals carried out by the DCU and by the TCMS will be positive, Incoherencies will be detected and stored as failures.

<"*Open DOORS*" Icon and "*porte_aperte.wav*" Audio >

Ta: door open warning time.



3.1.5.1.1 **Normal condition**

To check the coherence of the input and output signals, the DCU make available to TCMS as SW signals the state of the HW input signals and the TCMS make available to DCU as SW signal the required state of the hardwire TLs.

In particular the SW signal zero speed and V>=10 km/h shall not repeat the state of the corresponding TL, but shall give the effective state of the speed of the train, to allow detection of a failure state of TLs.

This is obtained by a coherence check among the state of the HW signals:

Door release 1, Door release 2, Zero speed, Speed >=10 km/h

and

the corresponding SW signals.

The DCU is provided by a HW interlock allowing to feed the door motor for opening only if at least one release TL is active.

In addition, also the zero speed TL have to be active or both the release TLs have to be active to feed the door motor for opening. This avoids that the door becomes released because a single failure of the input signals or because an internal failure of DCU.

At powering up of the DCU:

if the door and the step are closed, they remain closed,

if the door is open, it close with pre-warning acoustic signal and obstacle detection function active.

if the step is open, it close when the door leaf is in closed and locked position, with obstacle detection function active.

3.1.5.1.2 **Failure case**

a) Failure of one release TLs,
b) Failure of zero speed TLs,
c) Failure of V>=10 km/h signal at low speed
d) Failure of V>=10 km/h signal at high speed

3.1.5.1.3 **Failure detection**

a) see § 3.1.1.1.3
b) The zero speed TL level is compared with the door release TL and with the V>=10 km/h TL. If the door release train lines are both active and the V>=10 km/h is inactive the zero speed TL will be active. If not the zero speed TL is taken as failed and the SW zero speed signal is used when the door functionality has to be changed from standstill to running (pre-warning time before closing, obstacle detection).
c) The V>=10 km/h TL level is compared with the zero speed TL. If the zero speed signal from TL is not detected as failed and the states of the lines are not coherent the V>=10 km/h TL is detected as not deactivated for speed ≤10 km/h.
d) The V>=10 km/h TL level is compared with the speed value by TCMS. If the V>=10 km/h TL level in not consistent with the speed value a failure event is detected by TCMS.

3.1.5.1.4 **Failure handling**

a) The system continues to operate correctly. Failure to be removed during maintenance.
b) The system continues to operate correctly. Failure to be removed during maintenance.
c) The internal emergency handle is not operable at standstill. The failure causes a hazard
d) The internal emergency handle is not de-activated when the train is running at speed >=10 km/h. The failure causes a hazard

3.1.5.2 **Requirements**

ID	2F_14.Zefiro-Europe.TRS.52
Description	Status of each door shall be shown on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.303	The DCU shall send the current door and step status to TCMS	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.502	The current door and step status shall be shown upon request on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.500	The DCU shall send the status of the input HW signals ”Door release 1”, “Door release 2”, “Zero speed”, “V>=10 km/h” to TCMS.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.503	The TCMS shall send to each DCU, as SW signals, the required status of the input HW signals ”Door release 1”, “Door release 2”, “Zero speed”, “V>=10 km/h” .	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.504	The DCU shall compare status of HW and SW input signals ”Door release 1”, “Door release 2”, “Zero speed”, “V>=10 km/h” for internal diagnostic purpose.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.505	The TCMS shall compare status of the input HW signal received from each DCU ”Door release 1”, “Door release 2”, “Zero speed”, “V>=10 km/h” with	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

	the status of the respective signal on train level and a diagnostic event shall be set if they differ and the door is not selected to be prevented to be opened.				

ID	2F_14.Zefiro-Europe.TRS.57
Description	There shall be one set of events stored locally in the DCUs and one set that shall be stored in the TDS.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.304	DCU shall have the capability to store events	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.305	DCU shall have the capability to output events	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.728	When a failure event is sent from DCU to TCMS, a diagnostic event shall be stored in TDS system giving the information about location of failed door and suggested recovery action.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_14.Zefiro-Europe.TRS.236
Description	The control system shall be designed in such a manner that at least two independent faults must occur at the same time before the respecitve door opens un-commanded if the door is not released.
Safety level	N/A

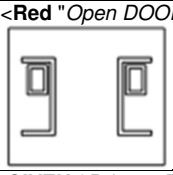
Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.306	In case of a fault causing one wrong input signal, the DCU shall not command the opening of the door, if the door is not released.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.307	One single internal fault of the DCU shall not cause the un-commanded opening of the door, if the door is not released.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

ID	2F_14.Zefiro-Europe.TRS.242
Description	After shut down, power failure, isolation or emergency operation, powering up of the door system shall be in a controlled manner to prevent injury.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.308	At powering up of the door system when the door and the step are closed, the door and the step shall remain closed. Further operation of the door shall be in accordance with the input signals.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.309	At powering up of the door system when the door is not closed, the door will close with audible warning and obstacle detection function active. Further operation of the door shall be in accordance with the input signals.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.310	With zero Speed active AND Doors Release 1 active OR Doors Release 2 active, at powering up of the door system when the step is not closed, the step shall close when the door is in closed and locked state. Further operation of the step shall be in accordance with the input signals.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

2F_14.Zefiro-Europe.CONCEPT.775	In case of Doors Release 1 and Doors Release 2 not active , at powering up of the door system, if the step is not closed, the step shall remains in the actual position and state of IFoot Step from DCU to TCMS shall be set to the value corresponding to state “Step cannot close”.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.776	With state of IFoot Step corresponding to state “Step cannot close”, train lines “Zero Speed” active and “Speed ≥10km/h” inactive, TCMS to inform Driver to set a door release command, in order to allow closing of the step.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_14.Zefiro-Europe.TRS.305
Description	Visualization of door status (vehicle green loop) is needed on IDU.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.795	**GIVEN** (Driver's Desk active) **WHEN** (Green Loop is de-energized (doors open)) **THEN** (Icon "*Open DOORS*" shall be represented on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.796	**GIVEN** (Driver's Desk active) **WHEN** (Green Loop is energized (doors closed and locked and steps retracted)) **THEN** (Icon "*Open DOORS*" shall be not represented anymore on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.798	<**Red** "*Open DOORS*" Icon> 	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.799	**GIVEN** (Driver's Desk active) **WHEN** (Green Loop is de-energized (doors open)) **THEN** (Audio file "*porte_aperte.wav*" shall be reproduced by IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.800	**GIVEN** (IDU is reproducing audio file "*porte_aperte.wav*") **WHEN** (Green Loop is energized (doors closed)) **THEN** (IDU shall interrupt audio file "*porte_aperte.wav*" reproduction)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.818	**GIVEN** (Driver's Desk active) **WHEN** (MIO data that acquires Green Loop is invalid) **THEN** (TCMS shall not activated icon on IDU and reproduce audio file "*porte_aperte.wav*")	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.819	**GIVEN** (MIO data that acquires Green Loop is invalid)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

	THEN (TCSM shall set a diagnostic event for the driver and the maintainer to declare that the Function is not available anymore)				

3.1.6 Special functions for PRM door

3.1.6.1 Function Design

The PRM doors are operable for opening and closing by the crew by square key switches internal and external.

The PRM door is interlocked with the lift for the wheel chair user to allow opening and closing of the door only when the lift is in stowed position.

3.1.6.1.1 Normal conditon

The local opening of the PRM doors have to be enabled, by the combination of the state of the HW TLs:

Door release 1, Door release 2, Zero speed, Speed >=10 km/h,as for the standard doors (see 3.1.1).

In addition operation of the PRM door requires an additional active input HW signal “lift in stowed position” ” (for door movement in closing position)

When the door is released and the signal “lift in stowed position” is active the door can be locally opened and closed by a key switch, operable by the train staff.

In addition, the PRM door closes when a general closing command is given by the train staff, by means of the crew switch, or by the driver, by means of the close push buttons on the driver desk, if the HW signal “lift in stowed position” is active.

If the lift is not in stowed position when a general closing command is given, the PRM door remains open until the lift is stowed.

No signal lights are provided for local indication of key switch active for opening command.

Closing of the door is with pre-warning acoustic and visual signals.

~~Because use of the lift is intermittent, an automatic periodical check of the oil level of the hydraulic actuator has been foreseen in order to generate a diagnostic message for maintenance before reaching the condition of non-efficiency of the system~~.

~~To this purpose, the TCMS activates periodically, for example once a day, for a predetermined time and with the lift in stowed position, the input of the lift (110 Vdc on input I) which activates the diagnostic function and detects during such activation the status of two outputs of the G and H.~~

~~The combination of the state assumed by the two outputs gives the indication of the proper oil level or the need for maintenance.~~

3.1.6.1.2 Failure case

a) As for § 3.1.1.1.2
b) Stowed lift signal not available
c) Stowed lift signal available also when the lift is not on stowed position.
d) Ethernet connection between DCU and TCMS not active
~~e) Low oil level of lift~~

~~f) Failure of oil level sensor of the lift~~

~~g) Failure of DC/DC converter or other failure of the lift~~

h) Lift NOT OK

3.1.6.1.3 **Failure detection**

a) As for § 3.1.1.1.3

b) The door does not close when requested

c) Failure not detectable.

d) The condition is detected by TCMs and DCU.

~~e) Output G of lift low and output H of lift high.~~

~~f) Both outputs G and H of lift high~~

~~g) Both outputs G and H of lift low~~

h) Output 2 and 7 of lift low

i) Lift Output 7 high and output 2 low

3.1.6.1.4 **Failure handling**

a) As for § 3.1.1.1.4

b) The door cannot be close.

c) The door can be used under the responsibility of the train crew.

d) The door can be used under the responsibility of the train crew.

~~e) If there are not visible leakage of the hydraulic system the lift may be used, but it have to be checked at the next planned maintenance stop of the train.~~

~~f) Oil sensor of the lift to be replaced~~

g) ~~DC/DC converter, wiring and oil sensor of the lift to be checked and replaced or restored~~

h) TBD

3.1.6.2 **Requirements**

3.1.6.2.1 **PRM door opening**

ID	2F_14.Zefiro-Europe.TRS.229
Description	The counter clockwise rotation shall provoke the opening of the door.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.313	The DCU shall open the PRM door when the square key switch for the PRM door is operated in counter clockwise rotation and the door is released.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

3.1.6.2.2 **PRM door closing**

ID	2F_14.Zefiro-Europe.TRS.194
Description	PRM door shall only be possible to close when the PRM lift is in "stowed position".
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.314	The PRM door shall only be possible to close if the PRM lift is in "stowed position".	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.311	Status of the PRM lift "stowed position" shall be sent to the relevant DCU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_14.Zefiro-Europe.CONCEPT.539	DCU shall send to TCMS the current lift status received from PRM lift.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.540	The current lift status shall be shown upon request on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
~~2F_14.Zefiro-Europe.CONCEPT.768~~	~~The TCMS shall activate periodically, (e.g. once a day), for a predetermined time (e.g. 10 s) and with the lift in stowed position, the diagnostic input of the lift (110 Vdc on input I)~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~		~~SIL 0~~
~~2F_14.Zefiro-Europe.CONCEPT.770~~	~~If the output G of lift is low and output H of lift is high when the diagnostic input is activate a CBM diagnostic message shall be given requiring a visual check of the lift and maintenance at next planned maintenance stop of the train.~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~	~~EVENT~~	~~SIL 0~~
~~2F_14.Zefiro-Europe.CONCEPT.772~~	~~If both output G and H of lift are high or low when the diagnostic input is activate a diagnostic message shall be given requiring a corrective maintenance action.~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~	~~EVENT~~	~~SIL 0~~

ID	2F_14.Zefiro-Europe.TRS.230
Description	The PRM door shall be closed when the general closing command is issued.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.315	When the doors release trainlines get deactivated,, the PRM door shall be closed if the signal “lift in stowed position” is active	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

ID	2F_14.Zefiro-Europe.TRS.231
Description	The clockwise rotation of the PRM door RIC key shall close the PRM door.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.316	The DCU shall close the PRM door when the square key switch for the PRM door is operated in clockwise rotation if the signal “lift in stowed position” is active	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

3.2 Local Door Control

3.2.1 Cab doors

3.2.1.1 Function design

The cab doors are manually operated doors.

3.2.1.1.1 Normal condition

The state of the door is monitored by a contact closed, when the door is closed and locked, included into the green loop.
If the door is not closed and locked the contact is open and consequently the green loop is open.
A second contact give to TCMS the door state for diagnostic purpose.

3.2.1.1.2 Failure cases

a) The contact does not close when the door is closed and locked.

3.2.1.1.3 Failure detection

a) The state of the door is detected by the TCMS and may be check by the driver on the diagnostic display on the page showing the door state.

3.2.1.1.4 Failure handling

a) A sealed bypass switch is provided to bypass the contact in the green loop in case of failure of the contact

3.2.1.2 Requirements

ID	2F_14.Zefiro-Europe.TRS.245
Description	If a cab door is open the green loop shall be interrupted
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.317	When the cab door is not closed and locked the output signal “Safety relay” (free contact) operated by the door shall be open	Derived Requirement	4S_12.01.-.Drivers Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.552	When the cab door is closed and locked the output signal “Safety relay” (free contact) operated by the door shall be closed.	Derived Requirement	4S_12.01.-.Drivers Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.553	A sealed by pass switch operable by the driver shall allow to by pass the safety contact.	Derived Requirement	4S_12.01.-.Drivers Door Systems		SIL 2

2F_14.Zefiro-Europe.CONCEPT.554	The TCMs detects the state of each cab door by means of a contact open when the door is in closed and locked position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.555	The current driver door state shall be shown upon request on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.556	TCMS shall supervise the sealed by pass switch and an event shall be set if it is operated	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.2.2 Open external door

3.2.2.1 Function Design

Each standard passenger door is provided with an internal and an external open push button for local opening command of a released door.

3.2.2.1.1 Normal condition

Starting state: door released, door and step not emergency operated and not out of service.

A door can be opened by the local internal or external open push button only if released.

When a door is released and it is closed steady green lighting of the open push button is active and it is active for an open command of the door.

During the opening and closing the green lighting of the open push button is activated with a blinking frequency of 2Hz

The green lighting of the close push button is activated as long as door is in open end position (steady activated) or during the door close delay with prewarning is active (LEDs are blinking with a frequency of 2Hz).

In this condition the operation of the open push button starts the opening sequence. If the door is in closing phase, because a local closing command from the close push button or elapsing of the open hold time, the door stops the closing sequence and re-open.

The opening sequence consists of the following steps:

- ♣ Opening of the step,
- ♣ When the step is fully open the door leaf starts to open
- ♣ When the door is fully open the green lighting of the close push button is activated

3.2.2.1.2 Failure cases

a) Activation of the push button is not detected by DCU
b) Failed lighting of push button
c) The step does not open
d) The door leaf does not open

3.2.2.1.3 Failure detection

a) Failure not detectable automatically
b) Failure not detectable automatically
c) Failure detected by DCU. Diagnostic message sent to TCMS
d) Failure detected by DCU. Diagnostic message sent to TCMS

3.2.2.1.4 **Failure handling**

a) The door have to be isolated
b) The door remains operating without the visual signal of released door on the push button.
c) The step have to be isolated
d) The door have to be isolated

3.2.2.2 **Requirements**

ID	2F_14.Zefiro-Europe.TRS.70
Description	Passengers shall have the possibility to open a released standard door by using the push button from inside and outside
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.318	If the door is released and the door is not isolated, the operation of one of the local open push buttons shall start the opening sequence of the door.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.319	A local opening command shall override a closing command form the local close puhbutton or elapsing of open hold time.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

3.2.3 **Close external door**

3.2.3.1 **Function Design**

The doors close under the following kinds of commands:

- Local commands:
  - Closing by the local close push button,
  - Automatic closing after hold-open time,
  - Closing by service push button,

- Central closing commands:
  - deactivation of the door release TLs
  - de-activation of the zero-speed signal.

Each standard passenger door is provided with an internal close push button for local closing command of a released door.

3.2.3.1.1 Normal condition

Starting state: door released and open, door and step not out of service or emergency operated.

Closing by the local close push button.

Each standard passenger door is provided with an internal close push button.

When the door is fully open the local close push button is active and the ~~red~~ green lighting on the button is on.

The operation of the close push button starts the normal closing sequence (see § 3.1.2).

The green lighting of the close push button is activated as long as door is in open end position (steady activated) or during the door close delay with prewarning is active (LEDs are blinking with a frequency of 2Hz).

As the door start the closing sequence the opening push buttons of the doors are activated and green lighted.

As the close push button is released it is de-activated and its red light turned off.

3.2.3.1.2 Automatic closing after hold-open time

If the hold open time function is active, as soon as the door reaches the full open position the hold open time starts. Expiration of the hold open time starts the normal closing sequence (see § 3.1.2).

As the door start the closing sequence the opening push buttons of the doors are activated and the green lighting of the open push buttons are activated with a blinking frequency of 2Hz and the close push button is de-activated and its lights turned off.

3.2.3.1.3 Closing by service push button

By pushing the service push button, the door starts closing sequence immediately without warning buzzer activation.

3.2.3.1.3 Closing by deactivation of the door release TLs.

When the door release TLs are de-activated and the zero speed signal is still active the DCU starts the normal closing sequence (see § 3.1.2).

3.2.3.1.4 Closing by deactivation of the zero speed signal.

When the zero speed TL is de-activated DCU starts the following closing sequence with a fix pre-warning time of 2 s:

- ♣ The acoustic warning signal is activated,
- ♣ The visual warning signal is activated,
- ♣ The door leaf starts to close without delay,
- ♣ When the door is in closed and locked position, the step start to close and the acoustic warning signal is de-activated.
- ♣ When the step is closed the section of the green loop related to the door is closed,

3.2.3.1.1 **Failure cases**

a) Activation of the close push button is not detected by DCU
b) Failed lighting of push button
c) The buzzer for warning signal not active
d) The step does not close
e) The door leaf does not close

3.2.3.1.2 **Failure detection**

a) Failure not detectable
b) Failure not detectable
c) Failure not detectable
d) Failure detected by DCU. Diagnostic message sent to TCMS
e) Failure detected by DCU. Diagnostic message sent to TCMS

3.2.3.1.3 **Failure handling**

a. The door cannot be closed locally. The door closes under the general closing command
b. The door remains operating without the visual signal of released door on the push button
c. The door remains operating without the acoustic warning signal. This condition is safety related. The door have to be isolated.
d. The step has to be reset after removing the obstacle or isolated
e. The door has to be reset after removing the obstacle or isolated

3.2.3.2 **Requirements**

ID	2F_14.Zefiro-Europe.TRS.104
Description	Closing command of the local standard door shall be activated by passengers through push buttons locate inside the coach.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.831	If the door is fully open and the zero speed signal is active and the door is not isolated, the operation of the local close push button shall start the closing sequence of the door and the close PB green lighting shall be changed from steady to blinking. The green leds are activated during the door close delay with prewarning (LEDs are blinking with a frequency of 2Hz) and remain active (in blinking mode) until the door is in close end position	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.832	If the door is fully open and a close command is provided to DCU, the red warning lamp installed on door pillar shall be activated with prewarning (LEDs are blinking with a frequency of 2Hz) and remain	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

	active (in blinking mode) until the door is in close end position				
2F_14.Zefiro-Europe.CONCEPT.733	If the local closing command is from the local close pusch button or by elapsing of “door hold open time”, it shall be possible to adjust via SW parameter if: the step shall start to close when the door leaf is closes and locked and the audible closing warning shall last until the step is closed, or the step shall start to close when the release TLs are deactiviated and the audible closing warning shall last until the door leaf is closed and locked.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.734	The step shall not start to close before the door leaf is closed and locked	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2

ID	2F_14.Zefiro-Europe.TRS.45
Description	If the door release signal is present, the door shall start to close after an adjustable time from the opening. In this case this closing can be interrupted by a local opening command.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.327	If the door is released and the zero speed signal is active the DCU shall start the closing sequence after “Door hold open time” from the opening.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.328	Via SW parameters, it shall be possible to adjust the “Door hold open time” or to deactivate the function.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.578	Operation of the open push button before the “Door hold open time” is expired shall reset the time counter “Door hold open time”	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.329	A local opening command shall override a closing command due to “Door hold open time” period expired	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.777	By pushing the service push button, the door starts closing sequence immediately without warning buzzer activation	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

ID	2F_14.Zefiro-Europe.TRS.218
Description	If door released is withdrawn, the door shall automatically close.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.330	When the release signals are both withdrawn, the DCU shall active the door closing sequence.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.579	If the door is in opening sequence, when the release signals are both withdrawn starts the closing sequence.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.580	When the release signals are both withdrawn, the local push buttons shall be de.activated.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

3.2.4 Door status indication

3.2.4.1 Function design

The door buttons are provided with lighting to indicate the active state of the button and to recognise that the button has been operated.

3.2.4.1.1 Normal condition

The open push buttons are green coloured, in accordance with EN 14752 requirement and when active are green lighted.
The open push button is active when the door is released and it is closed or in closing phase.
The close push button is red coloured, in accordance with EN 14752 requirement and when is active is green lighted.
The close push button is active when the door fully open.
~~When the active push buttons are operated green lighting is activated in addition to the green or red lighting.~~
Failure cases

a. See 3.2.2.1.2 b.
b. See 3.2.3.1.2 b.

3.2.4.1.2 Failure detection

a. See 3.2.2.1.3 b
b. See 3.2.3.1.3 b

3.2.4.1.3 Failure handling

a. See 3.2.2.1.4 b.
b. See 3.2.3.1.4 b.

3.2.4.2 Requirements

ID	2F_14.Zefiro-Europe.TRS.216
Description	The push button shall be illuminated when the door is released (possible to open).
Safety level	SIL 0

ID	2F_14.Zefiro-Europe.TRS.232
Description	An active push button shall give acknowledgement of its activation.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.834	The open PB green light shall be activated (steady) when the door is released and closed.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.835	During the opening sequence the green led Illumination of Open PB is activated with a blinking frequency of 2Hz.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.836	The open PB green lighting shall be deactivated when the door is in open end position	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.837	The close PB green lighting shall be activated when the door is in open end position (with the zero speed signal active and the door is not isolated).	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.838	The close PB green leds shall be deactivated when the door is in close end position	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

3.2.5 Local crew operation

3.2.5.1 Function Design

At each standard passenger door there is a control panel allowing the train staff to issue a general closing command of the other doors of the train, by the use of a crew switch operable by the standard square RIC key.
The manned door can be closed by the closing push button when the other doors are closed.
The closing command can be deleted, re-enabling the doors of the same side of the train, by operation of the same crew switch from which the general closing command had been issued.

3.2.5.1.1 Normal condition

Starting state: doors are released on one or both sides and one or more doors are open.
The closing command is available from the control panel of a released and fully open door.
The control panel at each door includes:

– A signal white lamp

– A crew switch with three operating positions:

- Neutral position: it is a stable position
- Closing position: rotating clockwise in respect to neutral position with spring return to neutral position
- Cancellation of closing command: rotating counter-clockwise in respect to neutral position with spring return to neutral position

The general closing command is issued by the clockwise operation of the crew switch.
The command applies to the doors on both sides of the train.
When the crew switch of a released and fully open door is operated clockwise the DCU send to TCMS a SW signal “Local panel closing command”.
The signal lasts for 5 s, independently how long the crew switch is kept in the closing position.

If the command is given during the pre-warning time before the door closing, the door stays open and the acoustic warning signal is switched off.
When the SW signal “close door” is received by TCMS, it de-activates all the door release TLs.
When the door release TLs are de-activated, the DCU of the not closed doors starts the normal closing sequence.
The DCU of the door where the closing command has been issued do not closes the door when the door release TLs are de-activated. Moreover, the local close push button and its illumination are deactivated.
When each door and the related step is closed and locked the related DCU send the TCMS a SW signal “door closed and locked”.
When the TCMS receives the SW signal “door closed and locked” from all doors, except from the door where the closing command has been issued, the TCMS send to the DCU of this door a SW signal “all other doors closed”.
When the DCU of the door where the closing signal has been issued receives the SW signal “all other doors closed”, it activates:

- the signal white lamp of the control panel
- the local closing push button, including the green illumination, for a configurable time (default value 5 s)

The door closes with the normal closing sequence by the operation of the local close push button within the time the push button is active.
If the door is not closed before elapsing of the time the close push button is active, the push button has to be re-activated for the configurable time, by turning in close position the crew switch.
When the door where the closing command has been issued closes, the DCU switches off the white signal lamp.
The closing command can be deleted by counter-clockwise rotation of the crew switch of the door where the closing command has been issued if the zero speed signal is active and the door is still open. This operation activates the release TLs only of the related train side, even if before the closing command the doors of both sides where released.
After restoring of the activation of the release TLs, all the doors of that train side are released and a new closing command can be issued from every released and open door o by the closing push button on the driver desk.

3.2.5.1.2 **Failure cases**

a. Operation of the crew switch is not detected by the DCU.
b. The Ethernet connection between DCU and TCMS is not active
c. The TCMS does not de-active one of the release TLs
d. The white lamp is not operating

3.2.5.1.3 **Failure detection**

a. The other doors do not close; the local close push button is not de-activated
b. Failure are detected by TCMS and by DCU; the other doors do not close
c. Failure are detected by TCMS; the other doors do not close

d. The white lamp remains off when the other doors are closed

3.2.5.1.4 **Failure handling**

a. To use another door for closing command issuing.
b. To use another door for closing command issuing if Ethernet connection is active.
c. To be open the supply switch of the control circuit of the related TL.
d. Closing of the other doors have to be checked visually.

3.2.5.2 **Requirements**

ID	2F_14.Zefiro-Europe.TRS.74
Description	The clockwise rotation of the crew switch shall provoke the closure of all doors excluding the door under control.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.331	When the crew switch is turned clockwise and the door is released and in open end postion the local DCU shall send a SW local panel closing command to the train control unit for 5 s stating from the operation of the crew switch and the door becomes the door under control.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.332	If the crew switch is operated during the prewarning time before door closing the prewarnig period is interrupted and the door under control shall remain open.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.333	When the train control unit receives a SW closing command from one DCU, the train control system shall deactivate the door release train lines related to both train sides.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.334	Deleted	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.335	The door under control shall not close automatically when door release signals are removed, because the deactivation of the release train lines.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.336	When the crew switch is operated in close position the local close push button and the green light of the local push button shall be de-activated until SW input signal “All other doors closed” is active.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

ID	2F_14.Zefiro-Europe.TRS.195
Description	
Safety level	SIL 2

The counter clockwise rotation of the crew switch shall provoke the cancellation of the general closing command previously given from the same door.

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.337	When the crew switch of the door under control is turned counter clockwise the DCU of the door under control shall activate an HW output “door enabling” interface.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.338	When the HW output “door enabling” interface is active the train control unit shall activate the release train lines of the related train side.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.339	The DCU shall no longer be in state 'door under control' when the 'door under control' is closed or when the door release train lines become active.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.340	After the cancellation of the general closing command from the door under control, a new closing command shall be possible from any released and open door of the train, or from the driver desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_14.Zefiro-Europe.TRS.198
Description	When the doors are closed by the crew switch the white lamp by the door shall be illuminated when all doors exluding the local door is closed. The indication lamp shall be turned off when also the local door is closed.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.720	Each DCU shall send to the train control system the SW signal “door and step closed and locked” when the door and the step are closed and locked.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.721	When all the door unless the door under control are closed the train control system shall activate the SW signal “only one door open” to the DCU of the door under control.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.722	When the DCU of the door under control receives the SW signal “only one door open” the white RIC light shall be activated by the DCU.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

ID	2F_14.Zefiro-Europe.TRS.196
Description	The door under local command shall close if the close push button is pushed within 5-15 s (configurable parameter) from switching on the white lamp.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.342	A local hold open time shall be started as soon as the DCU of the door under control receiives the signal “only one door open”.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.343	During the local hold open time, the close push button shall be active	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.840	When the close pushbutton of the door under control is active, the green light of the push button shall be activated.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.345	The door under control can be closed by operating on the active local close push button.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.346	The “Local hold open time” shall be possible to adjust via SW parameters,	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.841	When the close push button is released after the closing command, the green light of the push button shall be deactivated.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

2F_14.Zefiro-Europe.CONCEPT.609	When the door under control is closed and locked the DCU shall send to the train control system the SW signal “door and step closed and locked”.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.610	When also the door under control is closed the train control system shall de-activate the SW signal “only one door open”	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.611	When the door under control is closed the DCU shall de-acvtivated the white RIC light.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.613	When the local hold open time has expired and the door under controlis still open, the DCU shall deactive the close push button.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.842	If the door under control is open and the close pushbutton is not active, a closing command by the crew switch shall restart the hold open time and shall activate the close push button and the related green light.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

3.2.6 Status indication of each door

3.2.6.1 Function Design

The state of the door is indicated at each door by three indicator lights in a standard way for customer trains.

3.2.6.1.1 Normal condition

In the vestibule, above each door there are three state indicator lights:
Yellow light: Door system active and closed and not released
Green light: Door system active and released
White or red light (double colour indication): ~~Red~~ white only door or step is in cut out – ~~White~~ red both Door and step are isolated
In addition, the “door isolate” state is shown also on the external display on the door.
The yellow light is on when: the supply voltage is active, the DCU is not in failure condition, the door or the step are not isolated, the door is not emergency operated, the door and the step are in closed and locked position and is not released.
The green light is on when: the supply voltage is active, the DCU is not in failure condition, the door or the step are not isolated, the door is not emergency operated, the door is released.
The multicolour light has the ~~Red~~ white light on when: the supply voltage is active and the door or the step are isolated.
The multicolour light has the ~~White~~ red light on when: the supply voltage is active and both the door and the step are isolated.

3.2.6.1.2 Failure cases

a. The lights remain permanently off.
b. The lights remain permanently on

3.2.6.1.3 **Failure detection**

a. Failure not detectable
b. Failure not detectable

3.2.6.1.4 **Failure handling**

a. The failure does not affect the normal operation of the door. The failure has to be removed during maintenance.
b. The failure does not affect the normal operation of the door. The failure has to be removed during maintenance.

3.2.6.2 **Requirements**

ID	2F_14.Zefiro-Europe.TRS.146
Description	Near each door, inside the vestibules, the following lamps shall be present - Yellow, to indicate the status of door system active AND door closed AND not released - Green, to indicate the status of door system active AND released - White/red: to indicate the status of : door isolated or step isolated (~~Red~~ white light) door isolated and step isolated (~~White~~ red light)
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.354	When the door is active, closed and not released the DCU shall activate the local yellow light.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.355	When the door is active and released the DCU shall activate the local green light.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
	When the door or step is isolated the local white/red light shall be activate with ~~red~~ white light				SIL 0
2F_14.Zefiro-Europe.CONCEPT.356	When the door and step are isolated the local white/red light shall be activate with ~~white~~ red light	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

3.2.7 **Obstacle Detection**

3.2.7.1 **Function Design**

During movements of door leaf and step obstacle are detected both in close and open direction.

In case of obstacle detection, the door leaf and step stops the movements and after the stop the operation continues in a way depending from the direction of movements and from the train speed trying to reach the final normal position.

If obstacle is not removed the DCU send to TCMS a diagnostic message.

3.2.7.1.1 **Normal condition**

3.2.7.1.2 **Door obstacle detection**

Starting state: closing door or opening door.

During the closing movement of the door leaf obstacles are detected by the following means until the door reaches the closed position:

- by sensitive edge,
- by monitoring of motor current,
- by way/time monitoring.

If an obstacle is detected when the zero speed signals is active the door reopens. The close warning device stay active until the door is closed and locked or until the time corresponding to adjustable parameter “Switch off time for acoustic warning” is elapsed.
After reaching the fully open position the close delay time after reversing (adjustable parameter) is started. When the delay time after reversing is elapsed the door start to close.
After an adjustable number of closing attempts, the door stays fully open and a diagnostic code is generated and make available to TCMS.
The door can be closed manually or reactivation of normal operation is done by action on one of the local push buttons.

If an obstacle is detected when the zero speed signals is not active the door reverses approximately 100 mm in the first closing attempt. After 2 s. the door attempts to close again. If the obstacle is still detected, the door reverses again approximately 100 mm (second closing attempt). After 2 s the door attempts to close again. If the obstacle is still detected, the door continues closing and motor current control gets active. A diagnostic code is generated and made available to TCMS.
During the opening movement of the door leaf obstacles are detected by the following means until the door reaches the closed position:

- by monitoring of motor current,
- by way/time monitoring.

If the DCU detects an obstacle during the opening movements, the door is stopped.
After a delay of 2 s the door tries to continue the opening movement. If the obstacle is detected again, the door repeats this sequence.
After an adjustable number of successive opening attempts (default number: 3) the door stays in the current position and a diagnostic code is generated and make available to TCMS.
The DCU accepts this position as door open position

3.2.7.1.3 **Step obstacle detection**

During a closing sequence trapped obstacles are detected by the following means until the step reaches the closed position:

- Motor current monitoring
- Way/time monitoring

When an obstacle is detected the step reopens for approx. 30 mm upon activation of the obstacle detection. After that the closing cycle will be continued.

After three successive closing attempts (adjustable parameter) a diagnostic code is generated and the step is fully reopened.
The diagnostic code is transmitted by the DCU to TCMS.
The step can be closed manually or reactivation is done by action on one of the local push buttons.
During an opening sequence the step movement monitoring is done in the following ways:

- Motor current monitoring
- Way/time monitoring

Depending on the already reached opening width the following obstacle detection functions are distinguished:

- The step has not reached the minimum opening width:

The step recloses after activation of the obstacle detection approx. 20 mm. After that the opening sequence is continued.
After three successive opening attempts (adjustable parameter) before reaching the minimum opening width a diagnostic code is generated and the step get closed.
The diagnostic code is transmitted to the vehicle control unit.
The step can be moved manually or reactivation is done by action on one of the local push buttons or by activation or de-activation of the release TLs or Zero speed TL,
The door will also not be opened.

- The step has already reached or exceeded the minimum opening width (170mm):

The step recloses after activation of the obstruction detection approx. 20mm. The DCU accepts the current position as maximum open position.
There are no further opening attempts and no diagnostic code is generated.
Because of reaching the minimum opening width the opening sequence of the door will be started.

3.2.7.1.4 **Failure cases**

a. False obstacle detection.

3.2.7.1.5 **Failure detection**

The maximum number of attempts is reached

3.2.7.1.6 Failure handling

a. The door or step can be moved manually or reactivation is done by action on one of the local push buttons. If after reactivation the door or step is still wrongly operating, it has to be isolated.

3.2.7.2 Requirements

ID	2F_14.Zefiro-Europe.TRS.79
Description	The obstacle detection shall be active during closing and opening.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.357	Obstacle detection shall be active during the closing sequence of the door.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.358	Obstacle detection shall be active during the opening sequence of the door	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.632	Obstacle detection shall be active during the closing sequence of the step.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.633	Obstacle detection shall be active during the opening sequence of the step.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1

ID	2F_14.Zefiro-Europe.TRS.203
Description	During obstacle detection sequences the warning buzzer shall always be activated
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.359	During obstacle detection sequences the close warning buzzer shall be activated	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

ID	2F_14.Zefiro-Europe.TRS.201
Description	The door shall try to close a number of attempts before the door control stops any further door movements and sends an error message. Then the door shall be free to be pushed aside by hand.
Safety level	SIL 0

ID	2F_14.Zefiro-Europe.TRS.204
Description	Obstacle detection parameters shall be adjustable acting on door central unit.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.736	If during door closing sequence the obstruction detection gets active the door shall reopen	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.737	After reaching the fully open position the door shall closes when the “delay time after revesing” is elapsed.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

2F_14.Zefiro-Europe.CONCEPT.360	If the train speed is 0 km/h, after an adjustable number of closing attempts, the doors stays fully opened.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.738	The “delay time after revesing” shall be possible to adjust via SW parameters,	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.361	The number of door closing or opening attempts during obstacle detection shall be possible to adjust via SW parameters,	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.362	If the door fails to close after an adjustable number of closing attempts it shall be indicated to TCMS	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.634	When the DCU indicates that a door does no close a diagnostic event shall be displayed to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_14.Zefiro-Europe.CONCEPT.635	If the doors are ordered to close and train speed is >0km/h and an obstacle is detected the following shall happen: - the door shall have a short reversing, 10cm. . after 2s the door shall attempt to close . if an obstacle is still detected the door shall have a second short reversing, 10 cm. . after 2s the doors shall attempt to close. . if an obstacle is still detected the door shall be closed with reduced power.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.739	The acoustic warnig signal for for closing shall remain active until the door closes or a delay time is expired after the max number of closing attempts.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.740	The delay time of lasting of the warning acoustic signal after the max number of closing attempts shall be possible to adjust via SW parameters,	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.741	If during opening sequence the obstruction detection gets active the door shall stop the opening movement	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.742	After a delay of 2 s the door shall try to continue the opening sequence	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.744	After three successive opening attempts the door shall stay in current position	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.745	If the door fails to open after an adjustable number of opening attempts it shall be indicated to TCMS	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.746	When the DCU indicates that the door does no fully open a diagnostic event shall be displayed to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_14.Zefiro-Europe.CONCEPT.747	If during step opening sequence the obstruction detection gets active and the step has not reached the minumum opening width, the step shall reclose approx. 20 mm. After that the opening sequence is continued.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.748	After three successive opening attempts the step shall close	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.749	When the step close after reaching the max number of opening attempts, it shall be indicated to TCMS	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.750	When the DCU indicates that the step does not fully open or the current position is not accepted as maximum open position a diagnostic event shall be displayed to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_14.Zefiro-Europe.CONCEPT.751	If during step opening sequence the obstruction detection gets active and the step has reached or exceeded the minumum opening width (170 mm),	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1

	the step shall reclose approx. 20 mm and accepts the current position as maximum open position				
2F_14.Zefiro-Europe.CONCEPT.752	If the step is not fully open or the current position is not accepted as maximum open position the door leaft shall not open	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.753	If during step closing sequence the obstruction detection gets active and the step shall reopen approx. 30 mm. After that the closing sequence is continued.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 1
2F_14.Zefiro-Europe.CONCEPT.754	After three successive closing attempts the step shall get fully open	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.755	When the step fully open after reaching the max number of opening attempts, it shall be indicated to TCMS	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.756	When the DCU indicates that the step does no close a diagnostic event shall be displayed to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_14.Zefiro-Europe.CONCEPT.757	The number of step closing or opening attempts during obstacle detection shall be possible to adjust via SW parameters,	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

3.2.8 Isolation of Doors

3.2.8.1 Function Design

The door can be put out of service by a door lock out device from inside and from outside the vehicle.

Moreover, also the step can be put out of service by a door lock out device from inside the vehicle.

In case of station with short platform it is possible to deactivate the doors of the end coaches to prevent opening of door where the platform is not present. Selection of doors to de-activate is possible centrally by the train staff.

3.2.8.1.1 Normal conditions

3.2.8.1.1.1 Door lock out

The devices to isolate the door can be activated by a standard RIC square key.

Activation of the door lock out device under the condition that the leaf is closed causes:

- Mechanical locking of the door
- Inhibition of the motor circuit
- Closing of the step, if the power supply is available
- By pass of the door contacts of the local safety loop (except of contacts on the bottom locking units for the front and rear edge of the door leaf, detecting the effective locking state of the door after the isolation)
- Inhibition of the emergency opening devices
- Activation of the local visual state indication “Door isolated”
- Door status information of a diagnostic code made available to TCMS

The door lock-out device can be operated also when the door is not closed, but in any case the lock-out of the door and the bypass of the contacts become effective only when the door had been closed and locked

When the door lock-out device is activated the local contacts in the green loop related to the locked door are by-passed by a Door Lock Out Switch that is part of the lock out device.

3.2.8.1.1.2 Step lock out

The step can be put out of service by a step lock out device from outside the vehicle.
The step lock out device can be activated by a standard RIC square key.
Activation of the step lock out device under the condition that the step is closed causes:

- Mechanical locking of the step
- Inhibition of the motor circuit
- Closing of the door, if the power supply is available
- By pass of the step contacts of the local safety loop
- Step status information of a diagnostic code made available to TCMS

When the door lock-out device is activated the local contacts in the green loop related to the locked step are by-passed by a Step Lock Out Switch that is part of the lock out device.

3.2.8.1.1.3 Lock out indication

The status of door or step isolated is shown internally by the state indication white/red signal lamp with red light, near each door.
The status of door and step isolated is shown internally by the state indication white/red signal lamp with white light, near each door.
When a door will be put out of service the red leds lights of close and open pushbuttons will be switched on.
Moreover, the state of door out of service is shown outside the door by a pictogram on the display on the door

3.2.8.1.1.4 Selective door operation

A selective de-activation of the doors on the end cars is possible to prevent doors from opening.
The selection of doors to be de-activated is done centrally using the IDU, as well as if the de-activation of the selected doors is permanent until deselection, or a cab is activated.

3.2.8.1.2 Failure cases

a. By-pass of door contacts is not active
b. By-pass of step contact is not active
c. Ethernet connection between DCU and TCMS is not active

3.2.8.1.3 Failure detection

a. DCU detects the mechanical changeover of the by-pass contact and generate a status message to TCMS. If the contact is operated and the green loop remains open the failure is not detectable in a selective manner.

b. DCU detects the mechanical changeover of the by-pass contact and generate a status message to TCMS. If the contact is operated and the green loop remains open the failure is not detectable in a selective manner.

c. The door is not de-activated if centrally selected for de-activation

3.2.8.1.4 **Failure handling**

a. The green loop has to be by-passed by operation of the sealed switch.

b. The green loop has to be by-passed by operation of the sealed switch.

c. Failure of Ethernet connection is detected by TCMS and DCU. The related door has to be isolated locally.

3.2.8.2 **Requirements**

ID	2F_14.Zefiro-Europe.TRS.3
Description	It shall be possible to manually isolate one door from inside and outside, which electrically bypasses the green loop for this door, independent from the automatic lock.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.647	The door shall be possible to manually isolate from inside and outside	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.648	By taking the door isolated, it shall be mechanically locked in closed position. For that the door must be closed, or manually moved to closed position.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.649	When the isolating device is operated the door and step functions shall be switches off.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.650	When the door is isolated the door safety relay shall be bypassed.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.651	The DCU shall send to TCMS the states of door out of service and step out of service.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.652	The TCMS shall send to the PIS system the state of door out of service for visualisation of the state on the external display on the door.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.843	When a door will be put out of service the red leds lights of close and open pushbuttons will be switched on.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
	When the door is isolated TCMS shall : • show the door status isolated at IDU • filter (not consider) the door status data (open , close , emergency etc) • manage (consider) only the diagnostic data (major fault , fault information etc.)				

ID	2F_14.Zefiro-Europe.TRS.220
Description	The train shall have a Selective Door Operation function which, when operated, prevents doors on the end cars from opening.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
~~2F_14.Zefiro-Europe.CONCEPT.368~~	~~Corridor configuration: It shall be possible to select centrally the doors that are prevented to be opened.By selecting one coach, all doors in that coach shall be prevented to be opened on both sides. It shall be possible to select more than one coach, e.g. both DM cars.~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~	~~IDU~~	~~SIL 2~~
~~2F_14.Zefiro-Europe.CONCEPT.369~~	~~Corridor configuration: It shall be possible to select centrally if the prevention from opening is until: - the prevention is de-selected - new cab activation - next power off.~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~	~~IDU~~	~~SIL 2~~
2F_14.Zefiro-Europe.CONCEPT.653	**Corridor configuration:** When a door is selected by the "selective Door operation function" the door release input signals shall be removed from the DCU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.766	**Corridor configuration:** If selective door operation is active and the cab is activated an event shall be displayed to the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.2.9 Emergency opening exterior doors

3.2.9.1 Function Design

The standard passenger doors are provided with internal and external emergency opening device.

The internal opening device is operable only when the train speed is lower than 10 km/h.

If the power supply of the door system is active, when the door is emergency opened and the train is at standstill, also the step open automatically.

The PRM doors are provided with internal manual release device, operable only by the use of a square key and not intended for passenger use.

Emergency operation on an isolated door is not possible.

3.2.9.1.1 Normal condition

3.2.9.1.1.1 Internal emergency egress device

The internal emergency egress device is a two stage device with an engage in the first stage. It is provided with two square key devices.

A decoupling device with decoupling magnet prevents door emergency opening in case of speed ≥ 10km/h.

The flap is split into two levers, linked together with one of the square key device.

The first lever will be released by the square key device on the flap for train crew operation. On the second lever a breakable seal is included.

The decoupling magnet has to be active by Speed ≥10km/h to prevent door emergency opening during high speed.

An operated emergency egress device can only be reset by turning clockwise the lower square key device.

If the vehicle is running and the speed not lower than 10 km/h, the Speed >=10km/h TL is active and the decoupling magnet is directly energized by the high speed signal.

Pulling the emergency handle in this condition does not cause any emergency unlocking action. By releasing the handle returns back to the 15° locked position. The DCU transmits to the TCMS the SW signal “Door emergency opening activated” and the local warning buzzer is activated

If the Speed >=10km/h TL becomes not active, the decoupling magnet becomes deactivated too and it is necessary to pull the handle once again in the fully operated position. The door will be unlocked, the handle can be released and turns back to the 15° position and the door can be opened manually.

After resetting the emergency egress device, the door will stay in its current position until a signal from the local push buttons are set. Then normal door function is restored in accordance with the existing control signals.

3.2.9.1.1.2 External emergency access device

The external emergency access device is a single stage device and has the same function as the emergency egress handle.

The only difference is that the operation of the emergency access device is possible by the use of square key and return of the handle in normal position is by spring whenever it is released.

For operating the emergency access device, the square key has to be operated and the handle pulled.

Releasing the emergency handle the emergency device will be blocked automatically, without operating the square key.

By operating the emergency handle the emergency device on the drive unit is operated by means of a bowden cable and therefore a manual opening of the door is possible. Additionally, an electrical limit switch is actuated.

The door has to be re-activated using the local push buttons to restore the normal door functions.

Then normal door function, in accordance with the existing control signals is activated.

3.2.9.1.1.3 Manual release device inside

The Manual release device inside of the PRM door is a single stage device and has the same function as the emergency egress handle of the standard doors.

A decoupling device is provided with a decoupling magnet to be active by Speed >=10km/h to prevent door manual release opening during high speed.

The operation of the Manual release device inside is possible only by the use of square key and return of the handle in normal position is by spring whenever it is released.

Releasing the manual release device inside the device will be blocked automatically, without operating the square key.

After releasing, the manual release device inside, the door stays in its current position until a signal from the local open/close RIC key operated crew switch is set. Then normal door function is restored in accordance with the existing control signals.

3.2.9.1.2 Failure cases

a. The V>=10 km/h TL is not active when the speed is ≥10 km/h.

b. The decoupling magnet is not activated when the V>=10 km/h TL is active because internal failure
c. The V>=10 km/h TL remains active when the speed is <10 km/h.
d. The decoupling magnet remains active when the V >=10 km/h TL is not active

3.2.9.1.3 **Failure Detection**

a. The TCMS detects the state of the TL
b. Failure not automatically detectable.
c. The TCMS detects the state of the TL
d. Failure not automatically detectable.

3.2.9.1.4 **Failure handling**

a. The failure has to be removed during maintenance.
b. Check of decoupling magnet functionality to be carried out periodically
c. The failure has to be removed during maintenance.
d. Check of decoupling magnet functionality to be carried out periodically.

3.2.9.2 **Requirements**

ID	2F_14.Zefiro-Europe.TRS.207
Description	It shall only be possible to operate the internal egress device at speed < 10 km/h.
Safety level	N/A

ID	2F_14.Zefiro-Europe.TRS.209
Description	For train speeds v≥10km/h the internal emergency opening shall be prevented.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.370	The speed >=10 km/h signal shall active when the speed is >=10km/h	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.371	The internal egress device of the passenger doors and the manual release device of the PRM doors shall only be possible to operate to open the door if the speed >=10 km/h. signal is not active	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.779	If zero speed and speed ≥10km/h signal are not active, the door starts to close when the emergency handle inside will be resetted.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.780	If the emergency handle inside is pulled and the “speed ≥10km/h” signal get active, then the door starts to close as soon as the emergency handle inside will be released.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2

ID	2F_14.Zefiro-Europe.TRS.93
Description	The operation of the emergency handle causes also the step extraction, if the power supply is available.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.372	In case of the emergency handle is operated and zero speed signal is active, the step shall be extracted if the power supply is available	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

ID	2F_14.Zefiro-Europe.TRS.208
Description	Through the RIC key it shall also be possible to open a door without removing the seal of the emergency egress device.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.373	The emergency egress device shall be possible to operate through square key without breaking the sealing.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.664	Operation to opend the door shall be possible only if the speed >=10 km/h. trainline is not active	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2

ID	2F_14.Zefiro-Europe.TRS.210
Description	The external emergency handle to open exterior door from outside shall work independently from train speed.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.374	The external emergency handle shall be possible to operate independently from the train speed	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2

ID	2F_14.Zefiro-Europe.TRS.211
Description	An activation of the emergency opening device shall result in: 1. Unlocking of the door system 2. The green loop shall be interrupted, 3. The door shall send an “emergency opened” status to train control 4. Energizing of the door motor shall be prevented 5. Permanent warning sound shall be activated (when power supply is available)
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.375	Activation of the emergency opening devices of the passenger doors and of the manual release device of the PRM doors shall mechanically unlock the door leaf	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2
2F_14.Zefiro-Europe.CONCEPT.671	Unlocking of the door leaf due to activation of the emergency devices or manual release devices shall interrupt the green loop.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 2

2F_14.Zefiro-Europe.CONCEPT.670	The DCU shall detect operation of the emergency opening devices or manual release devices and send a SW signal “Door emergency opening activated” to TCMS.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.669	When one emergency opening devices or manual release devices is operated, the related DCU shall keep the door motor deenegised until the normal operation of the door is restored.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.668	When one emergency opening device is operated and the power supply is available, the related DCU shall locally activate an acoustic warnig signal.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.667	After activation, the internal emergency device shall only be possible to reset by the RIC key	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.666	After activation, the external emergency device and the manual release devices shall be automatically reset when released.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.665	If zero speed signal is active, after reset of the emergency devices, operation of the local close or open button shall restore the normal operation of the door. If the release condition are fulfilled, by operation of the local open push button the door shall stay open or move to fully open position, by operation of the local close push button the door shall close. If the release conditions are not fulfilled and by operation of the close push button the door shall close	Derived Requirement	4S_12.-.Exterior Door Systems		N/A
2F_14.Zefiro-Europe.CONCEPT.758	If zero speed signal is active, after reset of the manual release device, operation of the local open/close RIC key operated crew switch shall restore the normal operation of the door. If the release conditon are fulfilled, by operation of the local open/close RIC key in counterclockwise direction, the door shall stay open or move to fully open position, by operation of the local open/close RIC key in clockwise direction, the door shall close. If the release condition are not fulfilled, and by operation of the local open/close RIC key in clockwise direction, the door shall close	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

ID	2F_14.Zefiro-Europe.TRS.212
Description	Emergency opening shall not be possible when door system is isolated, i.e. locked out of service.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.376	Operation of the emergency access or egress devices shall not unlock an isolated doors.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

ID	2F_14.Zefiro-Europe.TRS.238
Description	An emergency egress device that has been operated shall be indicated as follows: - to the train system (on board crew or control centre); - locally at the related door (visually and/or audible). A clear proof of operation of the device shall be provided.(e.g. break glass, lead seal, data recording.) The exact method shall be specified.
Safety level	N/A

ID	2F_14.Zefiro-Europe.TRS.241

Description	An emergency access device that has been operated shall be reported on the train system
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.377	When the emergency device is operated, an indication shall be sent to the vehicle control unit	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.378	When the emergency device is operated the local door warning buzzer shall be activated	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.379	When the emergency device is operated, a diagnostic event shall be displayed.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_14.Zefiro-Europe.CONCEPT.732	The operation of the internal emergancy device shall cause the breaking of a seal. Moreover after operation the device have to be reset by use of the RIC key	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0

ID	2F_14.Zefiro-Europe.TRS.240
Description	Under power supply failure condition the access device shall remain functional and allow access, provided that the door has not been isolated in service.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.381	The emergency access device operability shall not depend on power supply availability.	Derived Requirement	4S_12.-.Exterior Door Systems		N/A

3.2.10 Train unit coupling and uncoupling

3.2.10.1 Function Design

Coupling and uncoupling of the train can be carried out also during revenue service at intermediate stations if a double train is split to reach different end stations or vice versa.

In this case the door has to maintain its released state until the train units are at standstill.

3.2.10.1.1 Normal condition

In case of coupling of two train units the one with active cab should have the door not released, because it is the one moving for coupling, while the passive one may have the door released or not released.

After coupling the doors state of the two coupled trains will be:

- not released if before coupling the doors of both trains were not released,
- released if before coupling the doors of one train were released.

In case of uncoupling of two coupled train units, after uncoupling the doors state of each train will be the same before uncoupling.

3.2.10.1.2 Failure cases

a) Released/unreleased state of doors is not maintained after coupling/uncoupling operation
b) Released/unreleased state of doors is not the same on the coupled trains.

3.2.10.1.3 Failure detection

a) Failure not detectable
b) Failure detected by TCMS

3.2.10.1.4 Failure handling

a) To give a new releasing or closing command
b) To give a new releasing or closing command

3.2.10.2 Requirements

ID	2F_14.Zefiro-Europe.TRS.301
Description	Status of released door s shall be maintained after the trains have been coupled/decoupled
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.684	Doors shall be released after coupling if before coupling the doors of one train were released.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_14.Zefiro-Europe.CONCEPT.685	After uncoupling of two couple trains the door of the two trains shall maintain the previous release/not release state before the uncoupling	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_14.Zefiro-Europe.CONCEPT.383	Doors release 1	L	2F_14.Zefiro-Europe.CONCEPT.269 2F_14.Zefiro-Europe.CONCEPT.270 2F_14.Zefiro-Europe.CONCEPT.274 2F_14.Zefiro-Europe.CONCEPT.275 2F_14.Zefiro-Europe.CONCEPT.279 2F_14.Zefiro-Europe.CONCEPT.281 2F_14.Zefiro-Europe.CONCEPT.282 2F_14.Zefiro-Europe.CONCEPT.285 2F_14.Zefiro-Europe.CONCEPT.286 2F_14.Zefiro-Europe.CONCEPT.287 2F_14.Zefiro-Europe.CONCEPT.288 2F_14.Zefiro-Europe.CONCEPT.290

			2F_14.Zefiro-Europe.CONCEPT.291 2F_14.Zefiro-Europe.CONCEPT.292 2F_14.Zefiro-Europe.CONCEPT.293 2F_14.Zefiro-Europe.CONCEPT.294 2F_14.Zefiro-Europe.CONCEPT.295 2F_14.Zefiro-Europe.CONCEPT.301 2F_14.Zefiro-Europe.CONCEPT.315 2F_14.Zefiro-Europe.CONCEPT.318 2F_14.Zefiro-Europe.CONCEPT.327 2F_14.Zefiro-Europe.CONCEPT.330 2F_14.Zefiro-Europe.CONCEPT.334 2F_14.Zefiro-Europe.CONCEPT.335 2F_14.Zefiro-Europe.CONCEPT.338 2F_14.Zefiro-Europe.CONCEPT.339 2F_14.Zefiro-Europe.CONCEPT.500 2F_14.Zefiro-Europe.CONCEPT.579 2F_14.Zefiro-Europe.CONCEPT.580 2F_14.Zefiro-Europe.CONCEPT.665
2F_14.Zefiro-Europe.CONCEPT.716	Doors release 2	L	2F_14.Zefiro-Europe.CONCEPT.269 2F_14.Zefiro-Europe.CONCEPT.270 2F_14.Zefiro-Europe.CONCEPT.274 2F_14.Zefiro-Europe.CONCEPT.275 2F_14.Zefiro-Europe.CONCEPT.279 2F_14.Zefiro-Europe.CONCEPT.281 2F_14.Zefiro-Europe.CONCEPT.282 2F_14.Zefiro-Europe.CONCEPT.285 2F_14.Zefiro-Europe.CONCEPT.286 2F_14.Zefiro-Europe.CONCEPT.287 2F_14.Zefiro-Europe.CONCEPT.288 2F_14.Zefiro-Europe.CONCEPT.290 2F_14.Zefiro-Europe.CONCEPT.291 2F_14.Zefiro-Europe.CONCEPT.292 2F_14.Zefiro-Europe.CONCEPT.293 2F_14.Zefiro-Europe.CONCEPT.294 2F_14.Zefiro-Europe.CONCEPT.295 2F_14.Zefiro-Europe.CONCEPT.301 2F_14.Zefiro-Europe.CONCEPT.315 2F_14.Zefiro-Europe.CONCEPT.318 2F_14.Zefiro-Europe.CONCEPT.327 2F_14.Zefiro-Europe.CONCEPT.330 2F_14.Zefiro-Europe.CONCEPT.334 2F_14.Zefiro-Europe.CONCEPT.335 2F_14.Zefiro-Europe.CONCEPT.338 2F_14.Zefiro-Europe.CONCEPT.339 2F_14.Zefiro-Europe.CONCEPT.500 2F_14.Zefiro-Europe.CONCEPT.579 2F_14.Zefiro-Europe.CONCEPT.580 2F_14.Zefiro-Europe.CONCEPT.665
2F_14.Zefiro-Europe.CONCEPT.385	Zero speed	L	2F_14.Zefiro-Europe.CONCEPT.275 2F_14.Zefiro-Europe.CONCEPT.294 2F_14.Zefiro-Europe.CONCEPT.301 2F_14.Zefiro-Europe.CONCEPT.318 2F_14.Zefiro-Europe.CONCEPT.326 2F_14.Zefiro-Europe.CONCEPT.327 2F_14.Zefiro-Europe.CONCEPT.360 2F_14.Zefiro-Europe.CONCEPT.372 2F_14.Zefiro-Europe.CONCEPT.500
2F_14.Zefiro-Europe.CONCEPT.386	Speed >= 10 Km/h	L	2F_14.Zefiro-Europe.CONCEPT.275 2F_14.Zefiro-Europe.CONCEPT.370 2F_14.Zefiro-Europe.CONCEPT.371 2F_14.Zefiro-Europe.CONCEPT.500 2F_14.Zefiro-Europe.CONCEPT.664
2F_14.Zefiro-Europe.CONCEPT.387	Lift stowed (valid only for PRM door)	L	2F_14.Zefiro-Europe.CONCEPT.311 2F_14.Zefiro-Europe.CONCEPT.312 2F_14.Zefiro-Europe.CONCEPT.314 2F_14.Zefiro-Europe.CONCEPT.315 2F_14.Zefiro-Europe.CONCEPT.316

2F_14.Zefiro-Europe.CONCEPT.388	Coding Handicap door	L	

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_14.Zefiro-Europe.CONCEPT.401	Door enabling	L	2F_14.Zefiro-Europe.CONCEPT.337
2F_14.Zefiro-Europe.CONCEPT.403	Passenger door safety relay	L	2F_14.Zefiro-Europe.CONCEPT.296 2F_14.Zefiro-Europe.CONCEPT.297
2F_14.Zefiro-Europe.CONCEPT.686	Cab door safety relay	L	2F_14.Zefiro-Europe.CONCEPT.317 2F_14.Zefiro-Europe.CONCEPT.552
2F_14.Zefiro-Europe.CONCEPT.687	Green loop	L	2F_14.Zefiro-Europe.CONCEPT.297 2F_14.Zefiro-Europe.CONCEPT.298 2F_14.Zefiro-Europe.CONCEPT.299 2F_14.Zefiro-Europe.CONCEPT.374 2F_14.Zefiro-Europe.CONCEPT.537
2F_14.Zefiro-Europe.CONCEPT.688	Sealed green loop bypass switch	L	2F_14.Zefiro-Europe.CONCEPT.537 2F_14.Zefiro-Europe.CONCEPT.538
2F_14.Zefiro-Europe.CONCEPT.404	Sealed cab door safety relay bypass switch	L	2F_14.Zefiro-Europe.CONCEPT.553 2F_14.Zefiro-Europe.CONCEPT.554 2F_14.Zefiro-Europe.CONCEPT.555

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_14.Zefiro-Europe.CONCEPT.30			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_14.Zefiro-Europe.CONCEPT.406	Central close signal	L	2F_14.Zefiro-Europe.CONCEPT.290

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_14.Zefiro-Europe.CONCEPT.408	Local panel closing command	L	2F_14.Zefiro-Europe.CONCEPT.280 2F_14.Zefiro-Europe.CONCEPT.281 2F_14.Zefiro-Europe.CONCEPT.331 2F_14.Zefiro-Europe.CONCEPT.333 2F_14.Zefiro-Europe.CONCEPT.334
2F_14.Zefiro-Europe.CONCEPT.409	Door out of service (isolated)	L	2F_14.Zefiro-Europe.CONCEPT.651 2F_14.Zefiro-Europe.CONCEPT.652
2F_14.Zefiro-Europe.CONCEPT.410	Step out of service (isolated)	L	2F_14.Zefiro-Europe.CONCEPT.651
2F_14.Zefiro-Europe.CONCEPT.411	Current door status	A	2F_14.Zefiro-Europe.CONCEPT.303 2F_14.Zefiro-Europe.CONCEPT.502

2F_14.Zefiro-Europe.CONCEPT.689	Current step status	A	2F_14.Zefiro-Europe.CONCEPT.303 2F_14.Zefiro-Europe.CONCEPT.502
2F_14.Zefiro-Europe.CONCEPT.692	Release 1 input status	L	2F_14.Zefiro-Europe.CONCEPT.500 2F_14.Zefiro-Europe.CONCEPT.504 2F_14.Zefiro-Europe.CONCEPT.505
2F_14.Zefiro-Europe.CONCEPT.693	Release 2 input status	L	2F_14.Zefiro-Europe.CONCEPT.500 2F_14.Zefiro-Europe.CONCEPT.504 2F_14.Zefiro-Europe.CONCEPT.505
2F_14.Zefiro-Europe.CONCEPT.413	Door and step closed and locked	L	2F_14.Zefiro-Europe.CONCEPT.609 2F_14.Zefiro-Europe.CONCEPT.720
2F_14.Zefiro-Europe.CONCEPT.414	Zero speed input status	L	2F_14.Zefiro-Europe.CONCEPT.500 2F_14.Zefiro-Europe.CONCEPT.504 2F_14.Zefiro-Europe.CONCEPT.505
2F_14.Zefiro-Europe.CONCEPT.415	V>= 10 km/h input status	L	2F_14.Zefiro-Europe.CONCEPT.500 2F_14.Zefiro-Europe.CONCEPT.504 2F_14.Zefiro-Europe.CONCEPT.505
2F_14.Zefiro-Europe.CONCEPT.717	Local Panel door enabling command	L	2F_14.Zefiro-Europe.CONCEPT.337 2F_14.Zefiro-Europe.CONCEPT.338
2F_14.Zefiro-Europe.CONCEPT.718	Emergency device operated	L	2F_14.Zefiro-Europe.CONCEPT.377 2F_14.Zefiro-Europe.CONCEPT.670
2F_14.Zefiro-Europe.CONCEPT.416	Door obstable detection	L	2F_14.Zefiro-Europe.CONCEPT.362 2F_14.Zefiro-Europe.CONCEPT.634 2F_14.Zefiro-Europe.CONCEPT.635
2F_14.Zefiro-Europe.CONCEPT.417	Lift stowed (valid only for PRM door)	L	2F_14.Zefiro-Europe.CONCEPT.539

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_14.Zefiro-Europe.CONCEPT.418	SW Release 1 input signal	L	2F_14.Zefiro-Europe.CONCEPT.503 2F_14.Zefiro-Europe.CONCEPT.504 2F_14.Zefiro-Europe.CONCEPT.505
2F_14.Zefiro-Europe.CONCEPT.419	SW Release 2 input signal	L	2F_14.Zefiro-Europe.CONCEPT.503 2F_14.Zefiro-Europe.CONCEPT.504 2F_14.Zefiro-Europe.CONCEPT.505
2F_14.Zefiro-Europe.CONCEPT.420	SW Zero speed input signal	L	2F_14.Zefiro-Europe.CONCEPT.503 2F_14.Zefiro-Europe.CONCEPT.504 2F_14.Zefiro-Europe.CONCEPT.505
2F_14.Zefiro-Europe.CONCEPT.421	SW 10 Km/h speed input signal	L	2F_14.Zefiro-Europe.CONCEPT.370 2F_14.Zefiro-Europe.CONCEPT.371 2F_14.Zefiro-Europe.CONCEPT.503 2F_14.Zefiro-Europe.CONCEPT.504 2F_14.Zefiro-Europe.CONCEPT.505
2F_14.Zefiro-Europe.CONCEPT.422	Only one door open	L	2F_14.Zefiro-Europe.CONCEPT.336 2F_14.Zefiro-Europe.CONCEPT.342 2F_14.Zefiro-Europe.CONCEPT.610 2F_14.Zefiro-Europe.CONCEPT.721 2F_14.Zefiro-Europe.CONCEPT.722
~~2F_14.Zefiro-Europe.CONCEPT.423~~	~~Selective door operation~~	~~L~~	~~2F_14.Zefiro-Europe.CONCEPT.369~~

5 References

[1] EC01P006716B - TRD EXTERNAL DOORS SYSTEM Rev. 09

[2] EC01P003332B – TRD ADDENDUM EXTERNAL DOORS SYSTEM Rev. 00

[3] EC010019516 - TRD EXTERNAL DOORS SYSTEM Rev. 01